

Final Project Proposal (Week 6 Checkpoint)

Course: COMP6140001 – Data Mining

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1. Introduction

Customer churn, the rate at which subscribers terminate their service, poses a significant financial challenge to the telecommunications industry, resulting in **substantial revenue loss** and increased customer acquisition costs. Proactive churn prediction is essential for implementing timely retention strategies, which ultimately boost customer lifetime value. Existing research often addresses churn prediction using various machine learning techniques, such as **Logistic Regression** and **Random Forest** (Adejo & Orji, 2021) or complex neural networks on proprietary datasets.

Our project differentiates itself by:

- Conducting a rigorous comparison of 5 models: Logistic Regression, Decision Tree, Random Forest, XGBoost, and SVM by utilizing IBM Telco Customer Churn dataset, a standardized public benchmark.
- Explicitly addressing the challenge of **class imbalance** using the **SMOTE** technique.
- Prioritizing the **F1-score** and **ROC-AUC** for reliable performance evaluation.

2. Theoretical Basis

This project utilizes systematic **Knowledge Discovery in Databases (KDD)** process (selection, cleaning, transformation, mining, and interpretation) to ensure our work is repeatable. Because we are predicting a **yes/no outcome** (churn or no churn), the core task is **supervised classification**. We will train our models using historical data where the outcome is already known.

The project will conduct a comparative analysis of five distinct classification models:

- **Logistic Regression:** Simple, effective for binary outcomes, and provides interpretable results.
- **Decision Tree:** Easy to understand and visualize; split data based on feature values.
- **Random Forest:** An ensemble method that combines many decision trees to reduce overfitting and improve accuracy.
- **XGBoost:** A powerful boosting algorithm known for consistently strong high-performance in classification tasks.
- **Support Vector Machine (SVM):** Finds the optimal boundary between classes, particularly useful in complex feature spaces.

For evaluation, we're focusing on:

- **Precision:** How many of the predicted churners left?
- **Recall:** Did we catch most of the customers who really did churn?
- **F1-score:** A balanced measure that combines both precision and recall.
- **ROC-AUC:** How well each model separates the 2 classes across different thresholds.

3. Methods

Dataset: We use the IBM Telco Customer Churn dataset (publicly available on [Kaggle](#)), containing demographic, service, and account information for 7,043 customers.

Preprocessing: Convert categorical variables using one-hot encoding; Handle missing values (if any), Normalize numerical features for SVM and distance-based methods.

Modeling Pipeline:

1. Split data into training (70%) and testing (30%).
2. Apply SMOTE to training set to balance classes.
3. Train and tune five models.
4. Evaluate using F1-score, precision, recall, and ROC-AUC.

Implementation Tools: Python (scikit-learn, imbalanced-learn), Jupyter Notebook for experimentation

4. References

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